Image Recognition Tool using OpenCV

Keshav Dhamija[1], Himanshu Nager[2], Digvijay Solanki[3]

[1]Student, Department of Computer Science and Engineering, Panipat Institute of Engineering and Technology, Kurukshetra University, Kurukshetra, India.

[2]Student, Department of Computer Science and Engineering, Panipat Institute of Engineering and Technology, Kurukshetra University, Kurukshetra, India.

[3]Student, Department of Computer Science and Engineering, Panipat Institute of Engineering and Technology, Kurukshetra University, Kurukshetra, India.

Abstract - The principle thought behind this undertaking is to build a tool that can recognize two different images of a person. This can be used to identify a lost person with their images. The search begins when a guardian/search team uploads an image of a lost person along with other details like height, body type, skin color, last seen on the web application. The application looks for a possible match in the database by looking in the particular category. If a match is found, the search team is notified. On the other hand, if someone finds a lost child, they can upload his/her image along with the other details specified already. The application will look in the database for a possible match and the search team will be notified.

Key Words: ***Image Recognition, Identification, Machine Learning, OpenCV, Artificial Intelligence.***

I. Introduction

It is estimated that a child goes missing in India every eight minutes. Many are trafficked as part of a nationwide trade that is separating children from their families. Millions end up in forced labour, domestic slavery and sex work, in what's become a lucrative industry. But despite this, it rarely dominates headlines, there's minuscule public or political outrage to do something about it.

Our web application provides a tool that can recognize a person by using OpenCV for image recognition on their picture.

II. Background

In the hour of Digital India, we have to live with the assistance of Information and Technology. No field must be left untouched by the internet. If something looks promising and has the potential to help the people, it must be developed quickly and put to work instantly. Hence, our tool. Information technology has greatly helped in crime investigation and minimizing crime. Our app can further help in finding the criminals by the location feature that can provide data about where most abductions are taking place and where more abducted people are being found.

The proposed framework has some center, improved and some extraordinary usefulness. Along these lines, how about we know them individually and utilization of this usefulness in this framework. To conquer the issue, we thought of the arrangement actualized as picture preparation of. Dlib and OpenCV libraries are used for image recognition.

The need for a tool like this was felt when traditional search methods started getting costlier and inefficient. The old method of searching for a lost person included printing thousands of 'lost person' posters and putting them up in major public places. But these methods had a very low success rate. Moreover, searching in more than one city is a challenging task let alone searching in an entire state.

III. Proposed System

The most important feature of the proposed system is that it can be used to identify a lost person with their images. The search begins when the image of a lost person is uploaded onto the web application by the search team along with other details like height, build, skin complexion and last seen. The application looks for a possible match in the database by looking in the particular category. If a match is found, the search team is notified. On the other hand, if someone finds a lost child, they can upload his/her image along with the other details specified already. The application will look in the database for a possible match and the search team will be notified.

The Image Recognition tool promises better results than the traditional ways as it substitutes the human factor in the image recognition process by artificial intelligence and the need to put up a large number of posters is also eliminated.

Features of the System:

- **Authentication:** Both parties have to create an account to be able to upload images onto the database. This prevents filling up the database with spam images.
- **Message Generation:** The app automatically sends a notification to both parties if a match is found.
- **User-Friendly Interface:** The interface of the web application is kept simple and straightforward to avoid any confusion.
- **Smart Updating of Database:** Once the person is found, their images are automatically deleted from the database to avoid processing time delay in other searches.

Benefits of the System:

- **Cost-Effective:** As it doesn't require putting up boards and posters, expenses like labour costs and material costs are eliminated.
- **Success Rate:** When the human element is substituted by artificial intelligence, the possibility of human error is eliminated.
- **Makes the Search Process Easier:** Rather than contacting every neighboring police station in the state, the app provides the ability to search through available records.
- **Makes the Search Process Faster:** Old methods used by the authorities were formulated at a time when technology was limited. This new method of using artificial intelligence makes the process faster and efficient.

IV. Working

Image Recognition and Identification tool is a platform that connects two groups of people. The ones that are looking for someone lost and the others that have found someone that is lost. The Image Recognition and Identification tool does its job by fastening the processes that are required for these two groups to meet.

Application Layout

On the front page, the app has two options.

- Lost Someone
- Found Someone

If the user selects the first option, they have to provide the details of the lost person like height, build, skin complexion, last seen along with a clear image. That image goes into the 'lost people' folder of the app database. In the back end, OpenCV does its job by comparing the image with every image available in the 'found person' folder. If a match is found. Both parties are notified.

If the user selects the second option, they have to provide the details of the found person like height, build, skin complexion with a clear image. That image goes into the 'found people' folder of the app database. In

the back end, OpenCV does its job by comparing the image with every image available in the 'lost person' folder. If a match is found. Both parties are notified.

Below is a diagram that explains how two parties use the Image Recognition Tool to look for a lost person 'Tom'.

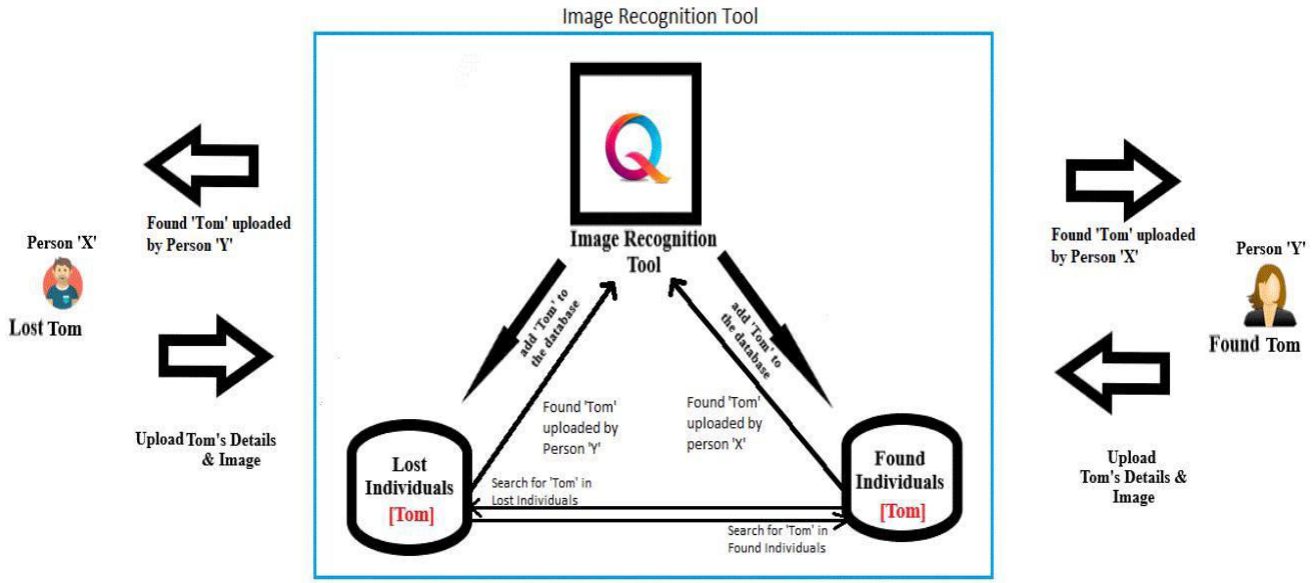

Fig. 1 Working of Image Recognition Tool

1. Person 'X' who happens to be Tom's guardian or a member of the search team that is looking for Tom logs into the Image Recognition Application and uploads Tom's details like age, height, sex, complexion along with an image of Tom.
2. That image goes into the 'Lost Individuals' folder of the app's database where it is compared with each entry in the 'Found Individuals' folder.
3. If someone (say Person 'Y') already found Tom and uploaded his details onto the app, OpenCV would identify that the Tom that Person 'X' is looking for is the same Tom that Person 'Y' uploaded in the 'Found Individuals' folder in the app's database and both parties would be sent a message notifying the receivers of Tom's whereabouts.

V. Conclusion and Future Scope

In the proposed system, we have implemented an image recognition system that would assist the search and rescue teams in the search for a missing person. The system could be used by authorities for several purposes like tracking down patterns of abductions and rescues, saving precious time by not opting for old methods, saving money by eliminating the labour and material costs of putting up posters, revolutionizing the search and rescue process by introducing an advanced tool, and paving the path for artificial intelligence to be used in police work.

The future scope for the proposed system is that it can be regularly updated with the addition of new features as the need arises. Integration with advanced machine learning and artificial intelligence would facilitate the introduction

of features like abduction heatmap and found location heatmap combined with other details like age, sex, and physical features which can be studied with machine learning to predict where the lost person could be found.

VI. References

- N Papa Rao, G Bhavana, M. Lakshmi Ravi Teja, "Image Recognition & Location Finder Using OpenCV", International Journal of Advanced Research in Computer Science Technology (IJARCST 2015).
- Maxim Mozgovoy, Evgeny Pyshkin, "Using Image Recognition for Testing Hand-drawn Graphic User Interfaces", International Journal of Scientific and Research Publications 2014.
- Alba Amato, Salvatore Venticinque, Beniamino Di Martino, "Image Recognition and Augmented Reality in Cultural Heritage Using OpenCV", International Journal of Advanced Research in Computer Science and Management Studies 2014.
- Guobo Xie, Wen Lu, "Image Edge Detection Based on Opencv", International Journal of Scientific and Research Publications 2013.
- Maxim Mozgovoy, Evgeny Pyshkin, "Using Image Recognition for Testing Hand-drawn Graphic User Interfaces", International Journal of Scientific and Research Publications 201